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A Cross-Layer SDN Orchestration Architecture for Stateful Edge Services: Design and Experimental Characterization

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Master's in Telecommunications and Computer Engineering

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Department of Information Science and Technology

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Acknowledgments

Write here your acknowledgements and, if applicable, any grants or sources of funding.

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Resumo

Palavras-Chave:

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Abstract

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Keywords: *Keywords; separated; by; semicolons*

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Figure 1.1 Global internet traffic between 2020 and 2025. Source: Statista, based on ITU data [1].

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List of Acronyms

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CHAPTER 1

Introduction

Orchestrating edge web services means dealing with coordination delays between routing/load balancing, auto-scaling and monitoring infrastructure. This has been shown to negatively impact the overall infrastructure and even QoS due to the unaccounted effects each module has on each other. In this chapter the context, motivation and problem statement are explained in 1.1 with the objectives presented in 1.2. The research questions that emerged are available in 1.3. The methodology used to develop the thesis is detailed in 1.4, the main contributions in 1.5 and finally the outlining of the rest of the 7 chapters of the thesis are presented in 1.6.

1.1. Context, Motivation, and Problem Statement

In the year 2025 global internet traffic surpassed 7 exabytes (1 exabyte = 1 000 000 000 gigabytes), this corresponds to a more than double increase in the amount of internet traffic generated since 2020 across fixed and mobile networks, this is represented in Figure 1.1. [1].

This growth has been driven by technological advancements that have in turn led to the explosion of real-time interactive and data-intensive applications. Examples include are web streaming and cloud gaming, which are services where users expect sub-second latency and high resolution multimedia. These are just a few examples that demonstrate the demand of infrastructure that is able to deliver performance with high availability and reliability at global scale, in other words regardless of where the end users are located.

Cloud computing is and has been the [2] backbone that enables these applications to operate at a global scale. It brings automatic scaling, both horizontally and vertically, a pay-per-use model and among other aspects, [2] a globally distributed offering and availability that has established it as the foundational layer that supports modern applications and services.

But at the same time, a growing percentage of global internet traffic comes from mobile devices, which now represent the main gateway to internet access [3]. This ongoing shift underscores that the majority of internet access comes from the network edge, far from the cloud data centers, evidencing the importance of edge infrastructure as a complement and or alternative to the cloud infrastructure for many scenarios. As presented in [4], [5] edge computing brings the computational resources to the network edge which and therefore closer to the end-users and end-products allowing for much lower latency in provisioning services and applications to users.

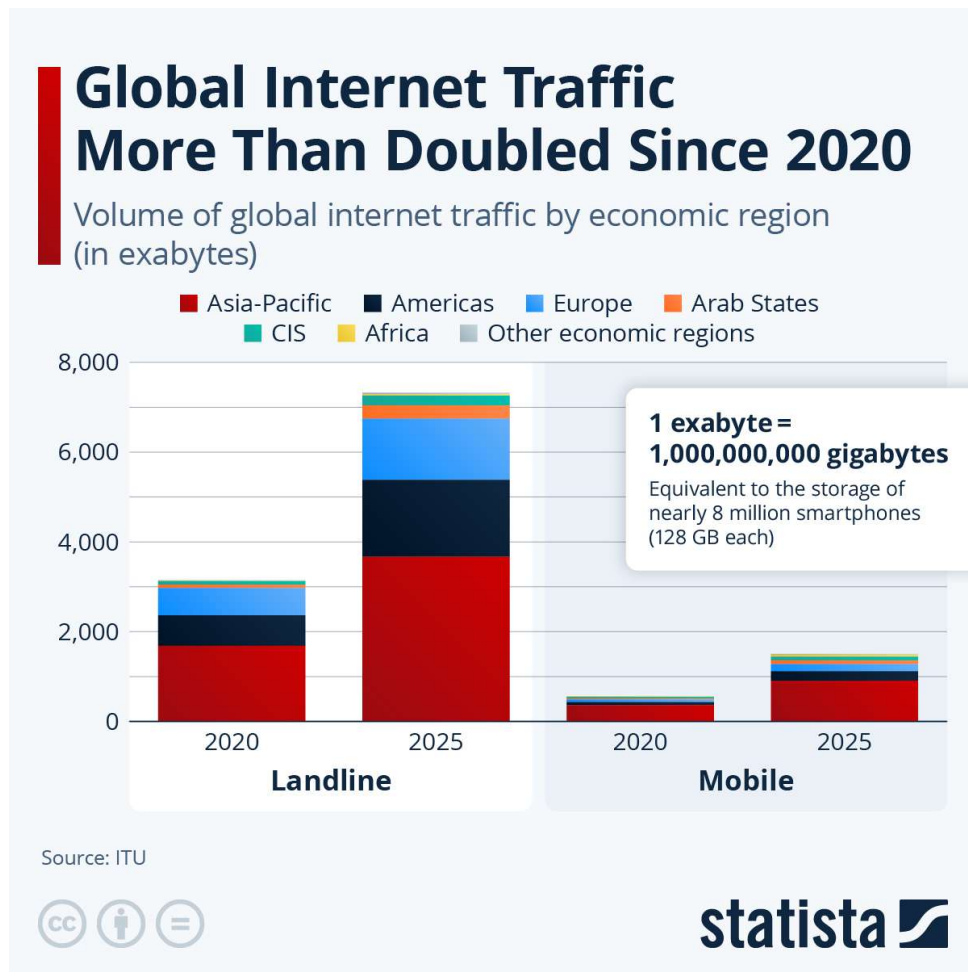


FIGURE 1.1. Global internet traffic between 2020 and 2025. Source: Statista, based on ITU data [1].

1.2. Objectives

1.3. Research Questions

1.4. Research Methodology

1.5. Main Contributions of this Dissertation

1.6. Dissertation Structure

CHAPTER 2

Background and Related Work

- 2.1. Review Methodology
- 2.2. The Unexamined Default: Three-Layer Separation
- 2.3. Auto Scaling
- 2.4. Load Balancing on SDN
- 2.5. Monitoring & Telemetry
- 2.6. Resource Orchestration on SDN
- 2.7. Summary and Research Gaps

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CHAPTER 3

Architecture and Design of Proposed Solution for Edge Services

This chapter presents the design of the proposed cross layer sdn orchestration architecture for edge services. It stems from the research gaps explored in the 2 and brings to the fold the need for an architecture that allows for each mentioned component to be independently managed and tested.

3.1. Design Requirements

3.2. Architecture

3.3. Distributed Elastic Allocation of Resources

3.3.1. Services

3.3.2. Data

3.4. Monitoring and Decision Engine

3.5. Control Workflow

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CHAPTER 4

Implementation of Proposed Solution

4.1. Experimental infrastructure

4.2. Distributed Elastic Allocation of Resources

4.2.1. Services

4.2.2. Data

4.3. Monitoring and Decision Engine

4.4. Control Workflow

4.5. Implementation Validation

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CHAPTER 5

Evaluation Methodology

5.1. Experimental Objectives

5.2. Evaluation Scenarios

5.3. Performance Metrics

5.4. Experimental Procedure with Results Statistical Analysis

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CHAPTER 6

Experimental Results

- 6.1. Trigger Quality — Overload Detection (RQ3)
- 6.2. Telemetry Freshness — Delivery Cadence (RQ1)
- 6.3. Backend Selection — Routing Awareness (RQ2)
- 6.4. Network Performance
- 6.5. Scalability Analysis
- 6.6. Discussion — The Compound Coordination Gap

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CHAPTER 7

Conclusions and Future Work

7.1. Conclusions

7.2. Research Contributions

7.3. Limitations

7.4. Future Research Directions

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